

WAFT Mac Shortcuts Research

Session Summary

January 11, 2026

WAFT Mac Shortcuts Research - Session Summary

Date: January 11, 2026

Session Type: Research Setup & PDF System Analysis

Status:  Complete

Session Overview

This session established the research infrastructure for "Example A: Mac Shortcuts Experiment" and analyzed the PDF generation systems in the WAFT codebase.

Work Completed

1. Research Folder Structure

Location: `WAFT-Mac-Shortcuts-Research/`

Created organized research structure:

- `README.md` - Main research document (Example A protocol)
- `documents/` - Research documents and content
- `notes/` - Analysis and observations
- `experiments/` - Experimental code (ready for future work)
- `WAFT_Research_Example_A_Mac_Shortcuts.pdf` - Initial PDF documentation

2. Research Question Forms

Created 4 comprehensive research question forms:

1. Pain as Data (`research_questions_01_pain_as_data.md`)
2. 7 questions on theoretical foundation
3. Topics: pain measurement, thresholds, transferability, accommodation

2. Interface Accommodation

(`research_questions_02_interface_accommodation.md`)

- 10 questions on technical implementation
- Topics: ambiguity tolerance, multi-modal integration, direct execution

3. Meta-Cognition (`research_questions_03_meta_cognition.md`)

- 10 questions on self-awareness
- Topics: genuine meta-cognition, self-documentation, consciousness

4. Human-AI Symbiosis

(`research_questions_04_human_ai_symbiosis.md`)

- 10 questions on relationship dynamics
- Topics: symbiosis definition, trust, capability amplification, societal impact

Total: 37 unanswered research questions ready for investigation

3. PDF System Analysis

File: `notes/pdf_systems_analysis.md`

Discovered three PDF generation systems:

1. Template System (PRODUCTION)



2. WeasyPrint + HTML + Jinja2

3. Used by showcase examples

4. Beautiful, automatic formatting

5. Production-ready

2. Foundation V1 (PRODUCTION)

- FPDF2 with blocks

- Used by

`generate_documentation_pdfs.py`

- Had formatting bug (fixed)

3. Foundation V2 (EXPERIMENTAL)

- FPDF2 with better typography
- Not production-ready yet

4. PDF Formatting Fix

Issue: Poor PDF formatting in

`WAFT_Research_Example_A_Mac_Shortcuts.pdf`

Root Cause:

`TextBlock.render()` in

`foundation.py` used manual

word wrapping with

`pdf.text()` which doesn't

handle:

- Word wrapping within margins
- Line spacing properly
- Page breaks automatically
- Y positioning correctly

Fix Applied:

Changed to use

`pdf.multi_cell()`

which handles all formatting automatically.

Files Modified:

- `src/waft/foundation.py`
 - **TextBlock.render()** method

Result:
PDF now
formats
correctly
with proper
word
wrapping
and
spacing.

5. Research
Content
Created

- documents/
WAFT_Research_V
-
Full
booklet
content
(Title,
Foreword,
3
Chapters,
Appendix)
 - generate_resea
-
PDF
generation
script
(needs
review/
update)
-

**Key
Findings**

**PDF
Generation
Systems**

- Temp System is the sophisticated tooling - Uses Weas handle format autom
- Found V1 is for quick market convey - Simple but had bugs
- Found V2

is
exper
-
Bette
typog
but
not
ready

